Every unit interval graph has its edge set partitioned into two comparability graphs

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Editorial note. The counted unit is the article's own Theorem unit at source lines 281-282 together with the author's proof that opens with the bold header Proof. at source line 285 and closes with the square at source line 300; the statement environment ends at line 282 and only blank lines and a medskip lie between, so the pairing is the ordinary adjacent one.

Editorial note. The retained context at source lines 117-120 and 128-129 is the article's own description of a comparability graph and its own definitions of the parameters p(G) and c(G); the retained context at source lines 272-273 is its own definition of interval graphs and of unit interval graphs.

Editorial note. The source preamble cannot be retained as it stands: it loads rawfonts, psfrag, verbatim, tikz with several libraries, hyperref with the pagebackref option and a dvi driver option for graphicx, none of which the retained body spans use, and it redefines backref commands that only exist because of that hyperref option. The retained preamble file therefore carries, line for line, source line 2, source line 3, source line 4, source line 16, source line 17, the theorem declarations of source lines 20-30 and the author macro definitions of source lines 32-41; the only textual change anywhere is the removal of the dvipdfm driver option from the graphicx line, because the assembly forces the xelatex engine. No new macro, package or declaration was introduced.

Editorial note. No retained line contains a citation or a cross-reference: the sentences of the article that cite Gilmore and Hoffman, Gavril and Walter are deliberately not retained, so the material carries no bibliography and needs no external navigation pointer.

Editorial note. The article's theorems co-triangulated, gensplit, bipline, loglog and ub, its lemma rec and its proposition H9 are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Comparability graph of a partially ordered set. Given a partially ordered set, its comparability graph has the elements as vertices and an undirected edge between two vertices exactly when the corresponding elements are comparable; the article's own description is retained at source lines 117-120.. Fixes the class of graphs into which the proof must partition the edge set of the interval graph, namely two edge-disjoint comparability subgraphs.

- The partition number p(G). p(G) is the minimum m such that the edge set of G is the union of m edge-disjoint comparability subgraphs of G; retained at source line 128, together with the covering number c(G) at source line 129.. The counted theorem asserts p(G) <= 2, so the parameter and its edge-disjointness requirement must be fixed before the statement can be read.

- Interval graph and unit interval graph. A graph is an interval graph when it is the intersection graph of a system of closed intervals of a line, and a unit interval graph when the intervals all have length one; the article's own definitions are retained at source lines 272-273.. Identifies the graph G of the counted theorem with the intersection graph G_U of a system U of unit intervals, which is the object the proof partitions.

Here we focus on a well known family of perfect graphs, the *comparability graphs*, which can be obtained from a partially ordered set as follows. Vertices correspond to the elements of the partially ordered set, and two vertices are connected by an undirected edge if and only if the corresponding elements

Let $p(G)$ denote the minimum number m such that $E(G) = \cup_{i=1}^{m} E(G_i)$ where the G_i-s are edge-disjoint comparability subgraphs of G. Dropping the condition that the G_i-s are edge-disjoint, we define $c(G)$ as the smallest number of comparability subgraphs of G needed to *cover* $E(G)$.

A graph G is an *interval graph* if it is the intersection graph of a system of closed intervals of a line. If the intervals have length one, their intersection graph is a *unit interval graph*.

Theorem 1. *For unit interval graphs G, $p(G) \leq 2$.*

Proof. Let U be a system of unit intervals and partition them as follows. Let $I_1 = [a_1, b_1] \in U$ be the interval with leftmost right endpoint and define X_1 as the set of intervals of U containing b_1. If $I_1, \ldots, I_k$ and $X_1, \ldots, X_k$ are defined and there are further intervals, select from them the interval $I_{k+1} = [a_{k+1}, b_{k+1}]$ with the leftmost right endpoint. Define X_{k+1} as the set of intervals not in $\cup_{i=1}^{k} X_i$ and containing b_{k+1}. The procedure ends with $[a_m, b_m]$ and X_m when all intervals of U are partitioned.

The unit interval graph of U, G_U, can be partitioned into two comparability graphs as follows. Define H_1 as the edge set of G_U defined by intervals of the *same* X_i. Note that H_1 is a union of m vertex-disjoint complete graphs. Let H_2 be the set of remaining edges, i.e. whose intersecting intervals belong to *different* X_i-s. Because U contains unit intervals, edges of H_2 must represent intersecting intervals belonging to X_i and X_{i+1} for some i. Thus H_2 is a bipartite graph. Clearly H_1, H_2 are both comparability graphs, finishing the proof. □